



**UNIVERSITY OF SCIENCE AND TECHNOLOGY
OF SOUTHERN PHILIPPINES**

Alubijid | Balubal | Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

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College of <u>Information Technology and Computing</u> Department of <u>Information Technology</u>		<u>SYLLABUS</u> Course Title: Elective 4 – Track A (Cybersecurity) Course Code: IT415 Credits: 3.0 units (2 hours Lecture, 3 hours Lab)
USTP Vision A nationally-recognized Science and Technology (S&T) university providing the vital link between education and the economy USTP Mission - Bring the world of work (industry) into the actual higher education and training of the students; - Offer entrepreneurs of the opportunity to maximize their business potentials through a gamut of services from product conceptualization to commercialization; - Contribute significantly to the national development goals of food security and	Semester/Year: 1st / S.Y. 2025-2026 Class Schedules: (Day Time/Room) IT4R1_Track1– Lec (M 10-12PM/Online) Lab (Th 1-4PM/Cisco 09-309) IT4R2_Track1– Lec (Th 1-3PM/Online) Lab (W 1-4PM/Cisco 09-309) IT4R3_Track1– Lec (W11-1PM/Online) Lab (T 7-10AM/Cisco 09-309) IT4R4_Track1– Lec (M 10-12PM/51-102) Lab (F 1-4PM/Cisco 09-309) IT4R5_Track1– Lec (W 9-11AM/Online) Lab (T 10-1PM/Cisco 09-309) IT4R6_Track1– Lec (Th 8-10AM/51-305) Lab (W 7-10AM/Cisco 09-309) IT4R7_Track1– Lec (T 6-8PM/Online) Lab (W 10-1PM/Cisco 09-309)	Prerequisite(s): IT325 – Elective 2 Co-requisite(s): None
	Instructor: Al-Monte Vince M. Calo Email: vince.calo@ustp.edu.ph Mobile No.: 0977-361-0019 Sections: IT4R2_Track1, IT4R6_Track1 Instructor: John Benedict L. Bernardo Email: jbl.bernardo@ustp.edu.ph Mobile No.: N/A Sections: IT4R1_Track1, IT4R4_Track1 Instructor: Archie O. Pachica Email: archie.pachica@ustp.edu.ph Mobile No.: 0995-905-0503 Sections: IT4R3_Track1, IT4R5_Track1	Consultation Schedule: Mon & Tue 8:00AM – 11:00AM Bldg. Rm. No.: IT Faculty Office, ICT Building, 4th floor Office Phone No./Local: (088)-857-1739 local 1154



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energy sufficiency through technology solutions.

Program Educational Objectives:

PEO1: Proficient in the IT field and able to engage constantly in technological and professional advancement by pursuing a higher academic level and/or practicing quality improvement in their professional career or entrepreneurial endeavor;

PEO2: Competent in generating new ideas and innovations in Information Technology with more emphasis on Technopreneurship, management, IT solutions and the likes through research collaborations; and

PEO3: Leading practicing professionals in the field of Information Technology who can contribute significantly to human development, socio-economic transformation, and patriotic initiatives.

Program Outcomes:

a: Apply knowledge of computing, science, and mathematics in solving computing/IT-related problems

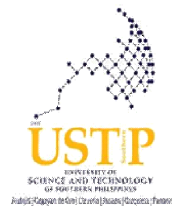
Instructor: **Ryan G. Sarip**
Email: **ryan.sarip@ustp.edu.ph**
Mobile No.: N/A
Sections: **IT4R7_Track1**

I. Course Description:

This course provides an in-depth exploration of cybersecurity fundamentals, focusing on the principles, technologies, and operations that defend modern networks. Learners will investigate the motivations behind cyber threats, the architecture of Windows and Linux operating systems, key networking protocols, and the role of security operations centers (SOCs) in cyber defense. The course offers hands-on exposure to tools and techniques used in threat monitoring, vulnerability assessment, malware analysis, digital forensics, and incident response. Real-world scenarios and case studies contextualize core concepts, while local and international regulations, such as GDPR and national cybersecurity frameworks, are examined to highlight the importance of compliance in cybersecurity operations. By the end of this course, students will be equipped with the foundational knowledge and skills to begin a career in cybersecurity or further their study in the field.

II. Course Outcome:

Course Outcomes (CO)	Program Outcome (PO)															
	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	
CO1: Identify and explain common cyber threats, vulnerabilities, and attack methods across networks, systems, and services.	I	E	E								E	E				
CO2: Apply basic security measures in Windows and Linux systems and use network tools for monitoring and analysis.	I	E	E				E					E	D			
CO3: Interpret security data and alerts to assess threats and support incident detection and response.		E	E				E						D			



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through critical and creative thinking.

b: Use current best practices and standards in solving complex computing/IT-related problems and requirements;

c: Analyze complex computing/IT-related problems by applying analytical and quantitative reasoning; and define the computing requirements appropriate to its solution;

d: Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer based-systems;

e: Design creatively, implement and evaluate different computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints;

f: Integrate effectively the IT-based solutions into the user environment with appropriate consideration for public health and safety, cultural, societal, and environmental concerns;

g: Select, adapt and apply appropriate techniques, resources, skills, and modern computing tools

CO4: Describe key cybersecurity policies, regulations, and defense strategies and their role in supporting security operations.

I E

III. Course Outline:

Allotted Time	Course Outcomes (CO)	Intended Learning Outcomes (ILO)	Topic/s	Suggested Readings	Teaching-Learning Activities	Assessment Tasks/Tools	Grading Criteria	Remarks
1 hr. Week 1 (Aug 11 - 16, 2025)		Course Orientation	- University's Vision and Mission - CITC Goals and Objectives - Class Policies and Agreement - Grading System - Course Requirements - Course Syllabus & Course Outline Presentation	- Student Handbook - Course Syllabus				
4 hrs. Week 1 (Aug 11-16, 2025)	CO1	- Explain why networks and data are attacked. - Explain how to prepare for a career in	Module 1 – The Danger - Introduction - War Stories - Threat Actors	<i>CCNA CyberOps Associate Curriculum</i> , Module 1-2	- Lecture/ Discussion - Lab Activities	Checkpoint Exam, Module 1-2		

to complex computing activities, with an understanding of the limitations;

h: Function effectively as individual, or work collaboratively and respectfully as a member or leader in diverse development teams and in multidisciplinary and/or multicultural settings;

i: Assist in the creation of an effective IT project plan;

j: Communicate effectively in both oral and in written form by being able to deliver and comprehend instructions clearly; and present persuasively to diverse audience the complex computing / IT-related ideas and perspectives;

k: Assess local and global impact of computing information technology on individuals, organizations, and society;

l: Act in recognition of professional, ethical, legal, security and social responsibilities in the utilization of information technology;

m: Recognize the need to engage in independent learning and be at pace with the latest developments in a specialized field in IT, with emphasis on Database Management

cybersecurity operations.

- Threat Impact
- Summary

Module 2 – Fighters in the War Against Cybercrime

- Introduction
- The Modern Security Operations Center
- Becoming a Defender

PIT

5 hrs.
Week 2
(Aug 18-23, 2025)

CO2

- Explain the security features of the Windows operating system.
- Implement basic Linux security.

Module 3 – The Windows Operating System

- Introduction
- Windows History
- Windows Architecture and Operations
- Windows Configuration and Monitoring
- Windows Security

Module 4 – Linux Overview

CCNA CyberOps Associate Curriculum, Module 3-4

- Lecture/ Discussion
- Lab Activities
- PIT

Checkpoint Exam, Module 3-4

and Information System; Network Design and Administration; and Computer Vision and Image processing for continual development as a computing professional; n: Participate in generation of new knowledge; or in research and development projects aligned to local and national development agenda or goals with the end view of contributing to the local and national economy; and o: Preserve and Promote “Filipino historical and cultural heritage”.				- Introduction - Linux Basics - Working in the Linux Shell - Linux Servers and Clients - Basic Server Administration - The Linux File System - Working with the Linux GUI - Working on a Linux Host					
	10 hrs. Week 3-4 (Aug 25-Sept 6, 2025)	CO1	- Explain how protocols enable network operations - Explain how the Ethernet and IP protocols support network communication.	Module 5 – Network Protocols - Introduction - Network Communication Process - Communications Protocols - Data Encapsulation Module 6 – Ethernet and Internet	CCNA CyberOps Associate Curriculum, Module 5-6	- Lecture/ Discussion - Lab Activities			

				- Introduction - Ethernet - IPv4 - IP Addressing Basics - Types of IPv4 Addresses - The Default Gateway - IPv6					
	5 hrs. Week 5 (Sept 8-13, 2025)	CO2, CO3	- Use ICMP connectivity verification tools. - Analyze address resolution protocol PDUs on a network.	Module 7 – Connectivity Verification - Introduction - ICMP - Ping and Traceroute Utilities Module 8 – Address Resolution Protocol - Introduction - MAC and IP - ARP - ARP Issues	CCNA CyberOps Associate Curriculum, Module 7-8	- Lecture/ Discussion - Lab Activities			

	5 hrs. Week 6 (Sept 15-20, 2025)	CO1	- Explain how transport layer protocols support network functionality. - Explain how network services enable network functionality.	Module 9 – The Transport Layer - Introduction - Transport Layer Characteristics - Transport Layer Session Establishment - Transport Layer Reliability Module 10 – Network Services - Introduction - DHCP - DNS - NAT - FTP and Sharing - Email - HTTP	<i>CCNA CyberOps Associate Curriculum, Module 9-10</i>	- Lecture/ Discussion - Lab Activities - PIT	Checkpoint Exam, Module 5-10		
	5 hrs. Week 7 (Sept 22-27, 2025)	CO1, CO2, CO4	Explain how network devices enable wired and wireless network communication.	Module 11 – Network Communication Devices - Introduction - Network Devices	<i>CCNA CyberOps Associate Curriculum, Module 11-12</i>	- Lecture/ Discussion - Lab Activities	Checkpoint Exam, Module 11-12		

			Explain how devices and services are used to enhance network security.	Wireless Communications Module 12 – Network Security Infrastructure - Introduction - Network Topologies - Security Devices		PIT			
	5 hrs. Week 8 (Sept 29- Oct 1, 2025)	CO1	- Explain how networks are attacked. - Explain the various types of threats and attacks.	Module 13 - Attackers and Their Tools - Introduction - Who is attacking our Network? - Threat Actors - Tools Module 14 – Common Threats and Attacks - Introduction - Malware - Common Network attacks - Using AI to analyze malware	<i>CCNA CyberOps Associate Curriculum, Module 13-14</i>	- Lecture/ Discussion - Lab Activities			
	Week 9 (Oct 2-8, 2025)	MIDTERM EXAM							
	5 hrs. Week 10 (Oct 13-18, 2025)	CO1, CO2, CO3	Explain network traffic monitoring.	Module 15 – Network Monitoring and Tools - Introduction - Network Monitoring	<i>CCNA CyberOps Associate Curriculum, Module 15-16</i>	Lecture/ Discussion			

			Explain how TCP/IP vulnerabilities enable network attacks.	- Tools Module 16 - Attacking the Foundation - Introduction - IP PDU Details - IP Vulnerabilities - TCP and UDP Vulnerabilities		Lab Activities			
	5 hrs. Week 11 (Oct 20-25, 2025)	CO1, CO4	- Explain how common network applications and services are vulnerable to attack. - Explain approaches to network security defense.	Module 17 - Attacking What We Do - Introduction - IP Services - Enterprise Services Module 18 – Understanding Defense - Introduction - Defense-in-depth Security Policies, Regulations, and Standards	<i>CCNA CyberOps Associate Curriculum, Module 17-18</i>	- Lecture/ Discussion - Lab Activities - PIT	Checkpoint Exam Module 13-17		
	5 hrs. Week 12 (Oct 27-Nov 1, 2025)	CO3, CO4	- Explain access control as a method of protecting a network. - Use various intelligence sources to locate current security threats.	Module 19 – Access Control - Introduction - Access Control Concepts - AAA Usage and Operation Module 20 – Threat Intelligence - Introduction	<i>CCNA CyberOps Associate Curriculum, Module 19-20</i>	- Lecture/ Discussion - Lab Activities - PIT	Checkpoint Exam Module 18-20		

				- Information Sources - Threat Intelligence Services						
	5 hrs. Week 13 (Nov 3-8, 2025)	CO2, CO3, CO4	- Explain how the public key infrastructure (PKI) supports network security. - Explain how a malware analysis website generates a malware analysis report.	Module 21 – Cryptography - Introduction - Integrity and Authenticity - Confidentiality - Public Key Cryptography - Authorities and the PKI Trust System - Applications and Impacts of Cryptography Module 22 - Endpoint Protection - Introduction - Antimalware Protection - Host-based Intrusion Prevention	CCNA CyberOps Associate Curriculum, Module 21-22	- Lecture/ Discussion - Lab Activities				
	5 hrs. Week 14 (Nov 10-15, 2025)	CO2, CO3, CO4	- Explain how endpoint vulnerabilities are assessed and managed. - Explain how security technologies	Module 23 - Endpoint Vulnerability Assessment - Introduction - Network and Server Profiling - Common Vulnerability Scoring System	CCNA CyberOps Associate Curriculum, Module 23-24	- Lecture/ Discussion - Lab Activities - PIT	Checkpoint Exam Module 21-23			

			affect security monitoring.	- Secure device Management - Information security Management Module 24 - Technologies and Protocols - Introduction - Monitoring Common Protocols - Security Technologies					
	5 hrs. Week 15 (Nov 17-22, 2025)	CO3	- Explain the types of network security data used in security monitoring. - Explain the process of evaluating alerts	Module 25 - Network Security Data - Introduction - Types of Security Data - End Device Logs - Network Logs Module 26 - Evaluating Alerts - Introduction - Sources of Alerts - Overview of Alert Evaluation	CCNA CyberOps Associate Curriculum, Module 25-26	- Lecture/ Discussion - Lab Activities - PIT	Checkpoint Exam Module 24-25		

	5 hrs. Week 16 (Nov 24-29, 2025)	CO3	- Interpret data to determine the source of an alert. - Explain how the CyberOps Associate responds to cybersecurity incidents.	Module 27 - Working with Network Security Data - Introduction - A Common Data Platform - Investigation Network Data - Enhancing the Work of the Cybersecurity Analyst Module 28 - Digital Forensics and Incident Analysis and Response - Introduction - Evidence Handling and Attack Attribution - The Cyber Kill Chain - The Diamond Model of Intrusion Analysis - Incident Response	CCNA CyberOps Associate Curriculum, Module 27-28	- Lecture/ Discussion - Lab Activities - PIT	Checkpoint Exam Module 26-28		
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	5 hrs. Week 17 (Dec 1-4, 2025)	CO4	- Describe key cybersecurity regulations (e.g., GDPR, HIPAA). - Explain how SOC's contribute to regulatory compliance.	Cybersecurity In the Philippines - National Cybersecurity Plan - Laws/Policies/Standards in force relating to cyber/information security. - Incident Response/Report an Incident - Awareness Campaign/Education/Events	National Cybersecurity Plan, 2023-2028	- Lecture/Discussion - Lab Activities	Reflection Paper		
	Week 18 (Dec 5-13, 2025)	FINAL EXAMINATION							

IV. Course Requirements:

1. Course Readings/Materials

a. Books, e-Books:

- CCNA CyberOps Associate Curriculum, available at netacad.com
- National Cybersecurity Plan, 2023-2028
- Cybersecurity Essentials, Brooks (2018) - *(Available at the USTP Library)*
- Security and Privacy in Cyber-Physical Systems - Foundations, Principles, and Applications, Song (2018) *(Available at the USTP Library)*
- Cybersecurity Law, Koseff (2017) *(Available at the USTP Library)*



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b. Supplies needed:

- CCNA Packet Tracer
- Wireshark
- Penetration Testing Tools
- Routers, Switches, Wireless Routers
- Network Cables (Serial, Console, UTP)
- Virtual Machine Software

c. URLs for Online Resources:

- <https://www.cisco.netacad.net>
- <https://www.nist.gov/cyberframework>
- <https://dict.gov.ph/national-cyber-security-plan?utm>
- <https://cms-cdn.e.gov.ph/DICT/pdf/NCSP-2023-2028-FINAL-DICT.pdf>

2. Class Attendance and Participation Policy

Reference: Student Handbook 2023 (Module 8. Attendance of Students)

Art. 1. Admission to Class

Enrolled students must show proof of their registration before being admitted to class. Upon the discretion of the instructor, students who are not enrolled may be allowed to attend a class as observer.

Art. 2. Tardiness and Absence

Prompt and regular attendance in classes is required of all students throughout the duration of every course. A student is considered tardy when s/he arrives past 25% of the scheduled class period (e.g., 30 minutes for a 1.5-hour period; 1 hour for a 3-hour period). Three (3) instances of tardiness are equivalent to one (1) absence. In cases where instructors/professors are late for class, students should not leave the classroom until the first third fraction of the scheduled class time has passed. Absences during the adding/dropping period shall not be included in the counting.

Art. 3. Excused Absences

It is the responsibility of the students to comply with class requirements in case of missed classes. Absences of students may be excused for any of the following reasons:

- Section 1. Illness

- Section 2. Natural calamities
- Section 3. Absences after midterm examination
- Section 4. Official participation in curricular or non-curricular Activities

Art. 4. Dropped Due to Absence

In compliance with BOR Resolution No. 31, s. 2018, the student who has incurred three (3) consecutive weeks of unexcused absences or more than seventeen percent (17%) of the total number of contact hours in the semester (e.g., 3 meetings for 3-hour classes/week) shall be automatically given a grade of Dropped with Failure (D/F) by the instructor. Once a student is given a midterm mark of D/F, the final grade shall automatically be D/F. A mark of "D/F" is not equivalent to a failing grade of 5.0.

3. Assignments, Assessment, and Evaluation

Grading System

Lecture Grade (67%)	
Criteria	%
Class Performance Item	10%
Quizzes	40%
Major Exams	30%
Performance Innovative Task / Project	20%
Laboratory Grade (33%)	
Criteria	%
Laboratory Exercises/Reports	30%
Laboratory Exam	40%
Hands-on Exercises	30%

Term/Periodic Grade: 67% Lecture Grade + 33% Laboratory Grade

Criteria and Weights in computing the Final Term Grade:

- If the FINAL EXAM coverage are all topics discussed from the beginning of the semester: $FG = 1/3 MTG + 2/3 FTG$

- If the FINAL EXAM coverage are all topics during the final term only: FG = 1/2 MTG + 1/2 FTG

Passing percentage: 70% (e.g., in a 10-item quiz, obtaining 7 pts would be equivalent to a passing score)

Rubric for PIT (Total: 100 points)

Criteria	Description	Points
Introduction and Background	Clarity of the project introduction and background information provided. Includes the context and motivation for the project.	15
Problem Statement and Objectives	Clear identification of the problem and well-defined project objectives. The objectives should be specific, measurable, achievable, relevant and time- bound (SMART)	20
Literature Review	Depth and relevance of the literature review, demonstrating knowledge of existing research and solutions related to the project topic.	20
Project Scope and Deliverables	Clear description of the project scope, including what will be achieved and what the deliverables will be.	15
Technical Approach and Methodology	Detailed explanation of the technical approach, including tools, technologies, and methodologies to be used. Feasibility and appropriateness of the proposed methods.	20
Timeline and Milestones	Realistic and detailed project timeline with key milestones and deadlines. Ensuring that the project is manageable within the given timeframe.	5
Formatting and Presentation	Adherence to proposal formatting guidelines	5

4. Use of VLE (USTeP) in class to distribute course materials, to communicate and collaborate online, to post grades, to submit assignments, and to give you online quizzes and activities.

Disclaimer: Every attempt is made to provide a complete syllabus that provides an accurate overview of the subject. However, circumstances and events make it necessary for the instructor to modify the syllabus during the semester. This may depend, in part, on the progress, needs, and experiences of the student.

Prepared by:

Recommending Approval:

Approved by:



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AL-MONTE VINCE M. CALO, DIT
Instructor

LOVE JHOYME M. RABOY, PhD
Department Chair

JUNAR A. LANDICHO, PhD
Dean